DeepSTARR - Publications

Format: Authors. Title. *Journal*. data;volume:pages. doi. PMID; PMCID. link

1. Foutadakis S, Bourika V, Styliara I, Koufargyris P, Safarika A, Karakike E.. Machine learning tools for deciphering the regulatory logic of enhancers in health and disease. *Front Genet*. Front Genet. 2025 Aug 13;16:1603687.. doi: 10.3389/fgene.2025.1603687. PMID: 40881170; PMCID: PMC12380740.
   Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12380740/`

   Note: Overview of AI methods

2. Zhou J, Rizzo K, Tang Z, Koo PK.. Uncertainty-aware genomic deep learning with knowledge distillation. *bioRxiv*. bioRxiv [Preprint]. 2024 Nov 15:2024.11.13.623485.. doi: 10.1101/2024.11.13.623485. PMID: 39605624; PMCID: PMC11601481.
   Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11601481/`

   Note: DEGU - framework for model ensembling (train with augmentations and predict mean): `https://github.com/zrcjessica/ensemble_distillation`

3. Duncan AG, Mitchell JA, Moses AM.. Improving the performance of supervised deep learning for regulatory genomics using phylogenetic augmentation. *Bioinformatics*. Bioinformatics. 2024 Mar 29;40(4):btae190.. doi: 10.1093/bioinformatics/btae190. PMID: 38588559; PMCID: PMC11042905.
   Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11042905/`

   Note: Homologs augmentation + Fine Tuning improve HS and DEV max of 0.03 (data leak?)

4. Yu Y, Muthukumar S, Koo PK.. EvoAug-TF: Extending evolution-inspired data augmentations for genomic deep learning to TensorFlow. *bioRxiv*. bioRxiv [Preprint]. 2024 Jan 18:2024.01.17.575961.. doi: 10.1101/2024.01.17.575961. PMID: 38293144; PMCID: PMC10827165.
   Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10827165/`

   Note: **EvoAug** - TensorFlow package for data augmentation, improve HS 0.03 and DEV 0.02

5. Lee NK, Tang Z, Toneyan S, Koo PK.. EvoAug: improving generalization and interpretability of genomic deep neural networks with evolution-inspired data augmentations. *Genome Biol.* Genome Biol. 2023 May 5;24(1):105.. doi: 10.1186/s13059-023-02941-w. PMID: 37143118; PMCID: PMC10161416.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10161416/`

Note: **EvoAug** - Pytorch package for data augmentation, improve HS 0.03 and DEV 0.02.

6. Seitz EE, McCandlish DM, Kinney JB, Koo PK.. Interpreting cis-regulatory mechanisms from genomic deep neural networks using surrogate models. *bioRxiv.* bioRxiv [Preprint]. 2024 Mar 2:2023.11.14.567120.. doi: 10.1101/2023.11.14.567120. PMID: 38013993; PMCID: PMC10680760.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10680760/`

Note: **SQUID** - model to predict focus of model (the lowest variance XAI): `https://github.com/evanseitz/squid-nn`.

7. Majdandzic A, Rajesh C, Koo PK.. Correcting gradient-based interpretations of deep neural networks for genomics. *Genome Biol.* Genome Biol. 2023 May 9;24(1):109.. doi: 10.1186/s13059-023-02956-3. PMID: 37161475; PMCID: PMC10169356.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10169356/`

Note: **Gradient Correction** - Improve XAI, angles histogram (hard math).

8. Li Z, Zhang Y, Peng B, Qin S, Zhang Q, Chen Y, Chen C, Bao Y, Zhu Y, Hong Y, Liu B, Liu Q, Xu L, Chen X, Ma X, Wang H, Xie L, Yao Y, Deng B, Li J, De B, Chen Y, Wang J, Li T, Liu R, Tang Z, Cao J, Zuo E, Mei C, Zhu F, Shao C, Wang G, Sun T, Wang N, Liu G, Ni JQ, Liu Y.. A novel interpretable deep learning-based computational framework designed synthetic enhancers with broad cross-species activity. *Nucleic Acids Res.* Nucleic Acids Res. 2024 Nov 27;52(21):13447-13468.. doi: 10.1093/nar/gkae912. PMID: 39420601; PMCID: PMC11602155.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11602155/`

Note: **CRE optimization** by genetic algorithm.

9. Song T, Song H, Pan Z, Gao Y, Dai H, Wang X.. DeepDualEnhancer: A Dual-Feature Input DNABert Based Deep Learning Method for Enhancer Recognition. *Int J Mol Sci.* Int J Mol Sci. 2024 Nov 1;25(21):11744.. doi: 10.3390/ijms252111744. PMID: 39519295; PMCID: PMC11546905.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11546905/`

Note: DeepDualEnhancer - Better Enhancer classification with BiLSTM.

10. Cheng W, Shen J, Khodak M, Ma J, Talwalkar A.. L2G: Repurposing Language Models for Genomics Tasks. *bioRxiv*. bioRxiv [Preprint]. 2024 Dec 10:2024.12.09.627422.. doi: 10.1101/2024.12.09.627422. PMID: 39713364; PMCID: PMC11661171.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11661171/`

Note: L2G - cross-modal transfer paradigm, beats HyenaDNA and BERTDNA.

11. Seitz EE, McCandlish DM, Kinney JB, Koo PK.. Uncovering the Mechanistic Landscape of Regulatory DNA with Deep Learning. *bioRxiv*. bioRxiv [Preprint]. 2025 Oct 8:2025.10.07.681052.. doi: 10.1101/2025.10.07.681052. PMID: 41278958; PMCID: PMC12632608.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12632608/`

Note: SEAM - Framework for gradient calculation.

12. Klie A, Laub D, Talwar JV, Stites H, Jores T, Solvason JJ, Farley EK, Carter H.. Predictive analyses of regulatory sequences with EUGENe. *Nat Comput Sci*. Nat Comput Sci. 2023 Nov;3(11):946-956.. doi: 10.1038/s43588-023-00544-w. PMID: 38177592; PMCID: PMC10768637.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10768637/`

Note: EUGENe - unified framework for genome predictions.

13. Reiter F, de Almeida BP, Stark A.. Enhancers display constrained sequence flexibility and context-specific modulation of motif function. *Genome Res*. Genome Res. 2023 Mar;33(3):346-358.. doi: 10.1101/gr.277246.122. PMID: 36941077; PMCID: PMC10078294.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10078294/`

Note: Motif's flanking has great impact.

14. Novakovsky G, Fornes O, Saraswat M, Mostafavi S, Wasserman WW.. ExplaiNN: interpretable and transparent neural networks for genomics. *Genome Biol*. Genome Biol. 2023 Jun 27;24(1):154.. doi: 10.1186/s13059-023-02985-y. PMID: 37370113; PMCID: PMC10303849.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10303849/`

Note: ExplainNN - Filters detect motifs (not really).

15. Woo HJ, Kim J, Kim SM, Kim D, Moon JY, Park D, Lee JS.. Context-dependent genomic locus effects on antibody production in recombinant Chinese hamster ovary cells generated through random integration. *Comput Struct Biotechnol J*. Comput Struct Biotechnol J. 2024 Apr 10;23:1654-1665.. doi: 10.1016/j.csbj.2024.04.023. PMID: 38680870; PMCID: PMC11046053.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11046053/`

Note: Biology - transgene integration site and composition of the rCHO cell.

16. Chen S, Loubiere V, Hollingsworth EW, Jacinto SH, Dizehchi A, Schreiber J, Kvon EZ, Stark A.. Predictive design of tissue-specific mammalian enhancers that function in vivo in the mouse embryo. *bioRxiv*. bioRxiv [Preprint]. 2025 Dec 24:2025.12.22.695948.. doi: 10.64898/2025.12.22.695948. PMID: 41497587; PMCID: PMC12767527.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12767527/`

Note: DeepSTARR model on mouse cells.

17. Santos M, Johari YB, Biggins L, Elliott NC, Schoenfelder S, Boddireddy M, Fabian DK, Anbar M, O'Callaghan PM, Rugg-Gunn PJ.. Comprehensive Mapping of Functional Enhancers in Chinese Hamster Ovary Cells. *Biotechnol Bioeng*. Biotechnol Bioeng. 2026 Jan;123(1):196-211.. doi: 10.1002/bit.70076. PMID: 41039989; PMCID: PMC12699134.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12699134/`

Note: DeepSTARR model on hamster cells.

18. Dalla-Torre H, Gonzalez L, Mendoza-Revilla J, Lopez Carranza N, Grzywaczewski AH, Oteri F, Dallago C, Trop E, de Almeida BP, Sirelkhatim H, Richard G, Skwark M, Beguir K, Lopez M, Pierrot T.. Nucleotide Transformer: building and evaluating robust foundation models for human genomics. *Nat Methods*. Nat Methods. 2025 Feb;22(2):287-297.. doi: 10.1038/s41592-024-02523-z. PMID: 39609566; PMCID: PMC11810778.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11810778/`

Note: Biology - Using DeepSTARR dataset.

19. Jeong CB, Cho H, Park D.. Deep learning application for genomic data analysis. *BMB Rep*. BMB Rep. 2026 Jan;59(1):60-68.. doi: 10.5483/BMBRep.2025-0020. PMID: 40962325; PMCID: PMC12867181.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12867181/`

Note: Overview AI model in genomics.

20. Cornejo-Páramo P, Zhang X, Louis L, Li Z, Yang Y, Wong ES.. Motif-based models accurately predict cell type-specific distal regulatory elements. *Nat Commun*. Nat Commun. 2025 Nov 24;16(1):10370.. doi: 10.1038/s41467-025-65362-2. PMID: 41285795; PMCID: PMC12644898.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12644898/`

Note: **BOM** - XGBoost on 0/1 motifs 0.78 on DEV and 0.42 in HS.

21. Rafi AM, Nogina D, Penzar D, Lee D, Lee D, Kim N, Kim S, Kim D, Shin Y, Kwak IY, Meshcheryakov G, Lando A, Zinkevich A, Kim BC, Lee J, Kang T, Vaishnav ED, Yadollahpour P; Random Promoter DREAM Challenge Consortium; Kim S, Albrecht J, Regev A, Gong W, Kulakovskiy IV, Meyer P, de Boer CG.. A community effort to optimize sequence-based deep learning models of gene regulation. *Nat Biotechnol*. Nat Biotechnol. 2025 Aug;43(8):1373-1383.. doi: 10.1038/s41587-024-02414-w. PMID: 39394483; PMCID: PMC12339383.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12339383/`

Note: **DREAM Challenge** - Prix Fixe framework to divide models into modular building blocks.

22. Chen X, Xu H, Yu S, Hu W, Zhang Z, Wang X, Yuan Y, Wang M, Chen L, Lin X, Hu Y, Cai P.. AI-Driven Transcriptome Prediction in Human Pathology: From Molecular Insights to Clinical Applications. *Biology (Basel)*. Biology (Basel). 2025 Jun 4;14(6):651.. doi: 10.3390/biology14060651. PMID: 40563902; PMCID: PMC12189417.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12189417/`

Note: Overview of AI methods

23. de Almeida BP, Schaub C, Pagani M, Secchia S, Furlong EEM, Stark A.. Targeted design of synthetic enhancers for selected tissues in the Drosophila embryo. *Nature*. Nature. 2024 Feb;626(7997):207-211.. doi: 10.1038/s41586-023-06905-9. PMID: 38086418; PMCID: PMC10830412.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10830412/`

Note: Transfer learning: `https://github.com/bernardo-de-almeida/DeepSTARR_embryo`

24. Song K, Ji H, Lee J, Yoon Y.. Microbial Transcription Factor-Based Biosensors: Innovations from Design to Applications in Synthetic Biology. *Biosensors (Basel)*. Biosensors (Basel). 2025 Mar 31;15(4):221.. doi: 10.3390/bios15040221. PMID: 40277535; PMCID: PMC12024804.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12024804/`

Note: Transcription factor-based biosensors, detecting TF in DeepSTARR data

25. D'Elia B, Fuxman Bass J.. Applications of high-throughput reporter assays to gene regulation studies. *Curr Opin Struct Biol*. Curr Opin Struct Biol. 2025 Oct;94:103105.. doi: 10.1016/j.sbi.2025.103105. PMID: 40580800; PMCID: PMC12221203.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12221203/`

Note: This article will be available in PMC on June 27, 2026

26. Fixsen BR, Han CZ, Zhou Y, Spann NJ, Saisan P, Shen Z, Balak C, Sakai M, Cobo I, Holtman IR, Warden AS, Ramirez G, Collier JG, Pasillas MP, Yu M, Hu R, Li B, Belhocine S, Gosselin D, Coufal NG, Ren B, Glass CK.. SALL1 enforces microglia-specific DNA binding and function of SMADs to establish microglia identity. *Nat Immunol.* Nat Immunol. 2023 Jul;24(7):1188-1199.. doi: 10.1038/s41590-023-01528-8. PMID: 37322178; PMCID: PMC10307637.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10307637/`

Note: Using DeepSTARR model for SALL1 gen.

27. Xu L, Liu Y.. Identification, Design, and Application of Noncoding Cis-Regulatory Elements. *Biomolecules.* Biomolecules. 2024 Aug 5;14(8):945.. doi: 10.3390/biom14080945. PMID: 39199333; PMCID: PMC11352686.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11352686/`

Note: Reference DeepSTARR on AI methods to detect TF.

28. Raicu AM, Fay JC, Rohner N, Zeitlinger J, Arnosti DN.. Off the deep end: What can deep learning do for the gene expression field?. *J Biol Chem.* J Biol Chem. 2023 Jan;299(1):102760.. doi: 10.1016/j.jbc.2022.102760. PMID: 36462664; PMCID: PMC9801099.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC9801099/`

29. Smith GD, Ching WH, Cornejo-Páramo P, Wong ES.. Decoding enhancer complexity with machine learning and high-throughput discovery. *Genome Biol.* Genome Biol. 2023 May 12;24(1):116.. doi: 10.1186/s13059-023-02955-4. PMID: 37173718; PMCID: PMC10176946.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10176946/`

Note: Discussion of AI methods.

30. Li X, Ye C, Liu T, Li S, Zhang M, Zhao Y, Jin Y, Cheng J, Yang G, Li P.. Engineering genetic elements for microbial protein expression systems: Advances, challenges, applications, and prospects. *Synth Syst Biotechnol.* Synth Syst Biotechnol. 2025 Oct 24;11:370-384.. doi: 10.1016/j.synbio.2025.10.008. PMID: 41280279; PMCID: PMC12629926.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12629926/`

Note: Overview of AI methods.

31. Ding N, Yuan Z, Ma Z, Wu Y, Yin L.. AI-Assisted Rational Design and Activity Prediction of Biological Elements for Optimizing Transcription-Factor-Based Biosensors. *Molecules.* Molecules. 2024 Jul 26;29(15):3512.. doi: 10.3390/molecules29153512. PMID: 39124917; PMCID: PMC11313831.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11313831/`

Note: AI methods in Activity Prediction.

32. Fishman V, Kuratov Y, Shmelev A, Petrov M, Penzar D, Shepelin D, Chekanov N, Kardymon O, Burtsev M.. GENA-LM: a family of open-source foundational DNA language models for long sequences. *Nucleic Acids Res.* Nucleic Acids Res. 2025 Jan 11;53(2):gkae1310.. doi: 10.1093/nar/gkae1310. PMID: 39817513; PMCID: PMC11734698.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11734698/`

Note: GENA-LM - Using DeepSTARR data to classify enhancers.

33. Chen Y, Zhang Y, Damoraki G, Wang S.. Editorial: Genetic horizons: exploring genetic biomarkers in therapy and evolution with the aid of artificial intelligence. *Front Genet.* Front Genet. 2026 Feb 6;17:1775149.. doi: 10.3389/fgene.2026.1775149. PMID: 41727388; PMCID: PMC12920033.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12920033/`

Note: Disscusion of synthetic enhancers.

34. Moeckel C, Mouratidis I, Chantzi N, Uzun Y, Georgakopoulos-Soares I.. Advances in computational and experimental approaches for deciphering transcriptional regulatory networks: Understanding the roles of cis-regulatory elements is essential, and recent research utilizing MPRAs, STARR-seq, CRISPR-Cas9, and machine learning has yielded valuable insights. *Bioessays.* Bioessays. 2024 Jul;46(7):e2300210.. doi: 10.1002/bies.202300210. PMID: 38715516; PMCID: PMC11444527.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11444527/`

Note: Mention about STARR-seq and other similar methods.

35. Zhang Z, Bao X, Jiang L, Luo X, Wang Y, Comai A, Waldhaus J, Hansen AS, Li W, Liu J.. Developing a general AI model for integrating diverse genomic modalities and comprehensive genomic knowledge. *bioRxiv.* bioRxiv [Preprint]. 2025 May 14:2025.05.08.652986.. doi: 10.1101/2025.05.08.652986. PMID: 40462903; PMCID: PMC12132192.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12132192/`

Note: Model on RNA assays.

36. Mulet-Lazaro R, Delwel R.. Oncogenic Enhancers in Leukemia. *Blood Cancer Discov.* Blood Cancer Discov. 2024 Sep 3;5(5):303-317.. doi: 10.1158/2643-3230.BCD-23-0211. PMID: 39093124; PMCID: PMC11369600.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11369600/`

Note: DeepSTARR as the SOTA model.

37. Loubiere V, de Almeida BP, Pagani M, Stark A.. Developmental and housekeeping transcriptional programs display distinct modes of enhancer-enhancer cooperativity in Drosophila. *Nat Commun*. Nat Commun. 2024 Oct 3;15(1):8584.. doi: 10.1038/s41467-024-52921-2. PMID: 39362902; PMCID: PMC11450171.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11450171/`

Note: Biological analyse of DeepSTARR data.

38. An Z, Jiang A, Chen J.. Toward understanding the role of genomic repeat elements in neurodegenerative diseases. *Neural Regen Res*. Neural Regen Res. 2025 Mar 1;20(3):646-659.. doi: 10.4103/NRR.NRR-D-23-01568. PMID: 38886931; PMCID: PMC11433896.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11433896/`

Note: DeepSTARR as the SOTA model.

39. Arnold M, Stengel KR.. Emerging insights into enhancer biology and function. *Transcription*. Transcription. 2023 Nov;14(1-2):68-87.. doi: 10.1080/21541264.2023.2222032. PMID: 37312570; PMCID: PMC10353330.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10353330/`

Note: DeepSTARR as the SOTA model and STARR-seq to detect enhancers.

40. Das M, Banerjee D, Hossain A, Jensen M, Mogre S, Sun J, Mao J, Glick AB, Salis HM, Girirajan S.. Quantifying the impact of genetic mutations on enhancer dynamics. *bioRxiv*. bioRxiv [Preprint]. 2025 Oct 21:2025.10.21.683558.. doi: 10.1101/2025.10.21.683558. PMID: 41279603; PMCID: PMC12633387.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12633387/`

Note: STARR-seq protocol.

41. Wang C, Lei B, Bao Y, Wang Z, Chen C, Zhang Y, Qin S, Sun T, Tang Z, Liu Y.. Multiomics analysis reveals critical cis-regulatory roles of transposable elements in livestock genomes. *iScience*. iScience. 2025 Feb 18;28(3):112049.. doi: 10.1016/j.isci.2025.112049. PMID: 40104067; PMCID: PMC11914811.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11914811/`

Note: DeepSTARR as the SOTA model.

42. Kuang Z, Yan X, Yuan Y, Wang R, Zhu H, Wang Y, Li J, Ye J, Yue H, Yang X.. Advances in stress-tolerance elements for microbial cell factories. *Synth Syst Biotechnol*. Synth Syst Biotechnol. 2024 Jun 28;9(4):793-808.. doi: 10.1016/j.synbio.2024.06.008. PMID: 39072145; PMCID: PMC11277822.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11277822/`

Note: DeepSTARR as the SOTA model.

43. Iurlaro M, Masoni F, Flyamer IM, Wirbelauer C, Iskar M, Burger L, Giorgetti L, Schübeler D.. Systematic assessment of ISWI subunits shows that NURF creates local accessibility for CTCF. *Nat Genet*. Nat Genet. 2024 Jun;56(6):1203-1212.. doi: 10.1038/s41588-024-01767-x. PMID: 38816647; PMCID: PMC11176080.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11176080/`

Note: Analyse TF in Catalytic activity of the imitation switch using DeepSTARR model.

44. Deng C, Whalen S, Steyert M, Ziffra R, Przytycki PF, Inoue F, Pereira DA, Capauto D, Norton S, Vaccarino FM, Pollen A, Nowakowski TJ, Ahituv N, Pollard KS.. Massively parallel characterization of psychiatric disorder-associated and cell-type-specific regulatory elements in the developing human cortex. *bioRxiv*. bioRxiv [Preprint]. 2023 Feb 16:2023.02.15.528663.. doi: 10.1101/2023.02.15.528663. PMID: 36824845; PMCID: PMC9949039.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC9949039/`

Note: Huge letniMPRA on human cortex data, DeepSTARR worse: no code.

45. Mulet-Lazaro R, Delwel R.. From Genotype to Phenotype: How Enhancers Control Gene Expression and Cell Identity in Hematopoiesis. *Hemasphere*. Hemasphere. 2023 Nov 8;7(11):e969.. doi: 10.1097/HS9.0000000000000969. PMID: 37953829; PMCID: PMC10635615.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10635615/`

Note: STARR-seq sequencing.

46. Zeitlinger J, Roy S, Ay F, Mathelier A, Medina-Rivera A, Mahony S, Sinha S, Ernst J.. Perspective on recent developments and challenges in regulatory and systems genomics. *Bioinform Adv*. Bioinform Adv. 2025 May 9;5(1):vbaf106.. doi: 10.1093/bioadv/vbaf106. PMID: 40626041; PMCID: PMC12233095.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC12233095/`

Note: DeepSTARR model as SOTA and STARR-seq as good technique.

47. Thomas HF, Buecker C.. What is an enhancer?. *Bioessays*. Bioessays. 2023 Oct;45(10):e2300044.. doi: 10.1002/bies.202300044. PMID: 37256273; PMCID: PMC11475577.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11475577/`

Note: STARR-seq to predict enhancers.

48. Lei S, Li C, She Y, Zhou S, Shi H, Chen R.. Roles of super enhancers and enhancer RNAs in skeletal muscle development and disease. *Cell Cycle*. Cell Cycle. 2023 Mar;22(5):495-505.. doi: 10.1080/15384101.2022.2129240. PMID: 36184878; PMCID: PMC9928468.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC9928468/`

Note: What is super-enhancers.

49. Mach P, Giorgetti L.. Integrative approaches to study enhancer-promoter communication. *Curr Opin Genet Dev*. Curr Opin Genet Dev. 2023 Jun;80:102052.. doi: 10.1016/j.gde.2023.102052. PMID: 37257410; PMCID: PMC10293802.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10293802/`

Note: Enhacer - Promoter interaction.

50. He AY, Palamuttam NP, Danko CG.. Training deep learning models on personalized genomic sequences improves variant effect prediction. *bioRxiv*. bioRxiv [Preprint]. 2025 Feb 15:2024.10.15.618510.. doi: 10.1101/2024.10.15.618510. PMID: 39463940; PMCID: PMC11507713.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11507713/`

Note: Enhancer design is emerging field.

51. Loell KJ, Friedman RZ, Myers CA, Corbo JC, Cohen BA, White MA.. Transcription factor interactions explain the context-dependent activity of CRX binding sites. *PLoS Comput Biol*. PLoS Comput Biol. 2024 Jan 16;20(1):e1011802.. doi: 10.1371/journal.pcbi.1011802. PMID: 38227575; PMCID: PMC10817189.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10817189/`

Note: DeepSTARR model as SOTA

52. Zeng X, Wei Z, Du Q, Li J, Xie Z, Wang X.. Unveil cis-acting combinatorial mRNA motifs by interpreting deep neural network. *Bioinformatics*. Bioinformatics. 2024 Jun 28;40(Suppl 1):i381-i389.. doi: 10.1093/bioinformatics/btae262. PMID: 38940172; PMCID: PMC11211823.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC11211823/`

Note: Deep-learning model is crucial for discovering motifs.

53. Penzar D, Nogina D, Noskova E, Zinkevich A, Meshcheryakov G, Lando A, Rafi AM, de Boer C, Kulakovskiy IV.. LegNet: a best-in-class deep learning model for short DNA regulatory regions. *Bioinformatics*. Bioinformatics. 2023 Aug 1;39(8):btad457.. doi: 10.1093/bioinformatics/btad457. PMID: 37490428; PMCID: PMC10400376.
Link: `https://pmc.ncbi.nlm.nih.gov/articles/PMC10400376/`

Note: **LegNET** as best model for short DNA regulatory regions.

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Note: DeepSTARR finding cooperative TFs

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Note: Motif insertions as ablation, just like DeepSTARR.

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Note: Use DeepShap as DeepSTARR.

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Note: Random changes in *loci*.

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Note: **ChromBPNet** - model to detect TF.